

INDIAN SPACE OLYMPIAD (ISO) 2026

SENIOR LEVEL - CLASSES IX-X

FIVE FULL-LENGTH MOCK QUESTION PAPERS

50 Questions Each | 250 Questions | Strictly Syllabus-Aligned

ASTRONOMY - SPACE SCIENCE - PHYSICS - REASONING

40 Core Questions + 10 Olympiad Challenge Questions per Paper

Practice Edition | Information cut-off: 23 July 2026

Indian Space School

Scope and Examination Design

This volume contains five independent 50-question mock papers for the Indian Space Olympiad Senior Level (Classes IX-X). Every question is restricted to the eleven topics in the supplied Senior Level syllabus. The language and mathematics are suitable for well-prepared Classes IX-X students, while the final ten questions in each paper require stronger multi-step reasoning, calculations or interpretation.

Pattern of Each Paper

Section	Questions	Suggested level	Marks
Core	1-40	Foundational to moderate Olympiad level	40
Olympiad Challenge	41-50	Difficult, multi-step and application-based	10
Total	50	Suggested time: 90 minutes	50

Suggested marking: 1 mark for each correct answer and no negative marking during practice. Schools may apply a different marking scheme if required.

Strict Syllabus Coverage

- 1. Fundamentals of Astronomy: celestial coordinates, distances, telescopes, electromagnetic spectrum, observational astronomy and basic calculations.
- 2. The Solar System: formation and evolution, geology, atmospheres, Kepler's Laws, orbital motion, small bodies and basic exoplanets.
- 3. Stars, Galaxies & the Universe: stellar evolution, the H-R diagram, stellar remnants, galaxies, Big Bang Theory and expansion.
- 4. Earth & Moon: Earth system, atmosphere, magnetosphere, climate, lunar geology, Earth observation and basic space weather.
- 5. Satellites & Space Technology: satellite systems, communication, navigation, remote sensing, spacecraft subsystems and applications.
- 6. Rockets & Launch Vehicles: propulsion, staging, escape velocity, orbital insertion, cryogenic engines and reusable vehicles.
- 7. ISRO & India's Space Missions: launch vehicles, Chandrayaan, Mars Orbiter Mission, Aditya-L1, Gaganyaan, XPoSat and future roadmap.
- 8. Space Exploration: planetary and deep-space missions, human spaceflight, stations, asteroid missions and future exploration.
- 9. Basic Physics for Space Science: mechanics, gravitation, circular motion, waves, optics, electricity, magnetism, thermal and basic modern physics.
- 10. Scientific Reasoning & Logical Aptitude: investigation, data analysis, mathematical reasoning, logic and application-based problem solving.
- 11. General Space Awareness: current astronomy, global missions, commercial spaceflight, policy and emerging technologies.

Common Constants and Relations

Quantity / relation	Value / expression
Speed of light, c	$3.0 \times 10^8 \text{ m/s}$
Earth radius, R_E	$6.4 \times 10^6 \text{ m}$
Standard gravity, g	9.8 m/s^2 ; use 10 m/s^2 when stated
1 parsec	3.26 light-years
Parallax relation	distance (pc) = $1 / \text{parallax (arcsecond)}$
Kepler relation for the Solar System	$T^2 = a^3$ when T is in years and a in AU
Circular orbital speed	v proportional to $1/\sqrt{r}$
Escape speed	$v_{\text{escape}} = \sqrt{2GM/R}$
Wave relation	$v = \text{frequency} \times \text{wavelength}$
Thin lens relation using magnitudes	$1/f = 1/v + 1/u$

MOCK PAPER 1

Senior Level (Classes IX-X) - 50 Questions - 50 Marks

Name		Roll Number	
School / Centre		Score	/ 50

Instructions: Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. Select the one best answer from A, B, C and D. Suggested time: 90 minutes.

SECTION A - CORE QUESTIONS

- Which coordinate remains nearly fixed for a distant star during one night?
 - Altitude
 - Azimuth
 - Hour angle
 - Declination
- A nearby star has annual parallax 0.25 arcsecond. Its distance is
 - 13.0 parsecs
 - 0.25 parsec
 - 40 parsecs
 - 4 parsecs
- A reflecting telescope forms its primary image mainly with a
 - glass prism
 - radio transmitter
 - flat window
 - curved mirror



Figure 1. Simplified electromagnetic spectrum.

- Study Figure 1. Which band lies immediately beyond visible light on the higher-frequency side?
 - Ultraviolet
 - Infrared
 - Microwave
 - Radio
- The Solar Nebula Theory states that the Solar System formed from a rotating cloud of
 - gas and dust
 - only liquid water
 - dark energy
 - solid iron already shaped as planets
- A planet with many old impact craters and little resurfacing is likely to have
 - rapid plate tectonics
 - a deep global ocean
 - constant erosion by rain
 - low recent geological activity
- According to Kepler's first law, a planet moves around the Sun in
 - an ellipse with the Sun at one focus
 - a circle with Earth at the centre
 - a straight line
 - an ellipse with the Sun at the centre
- A meteoroid that reaches Earth's surface is called a
 - meteorite
 - meteor
 - comet tail
 - dwarf planet
- A Sun-like star spends most of its life fusing hydrogen while it is on the
 - white-dwarf branch
 - event horizon
 - asteroid belt
 - main sequence
- On a Hertzsprung-Russell diagram, the main sequence runs roughly from
 - hot luminous stars to cool faint stars
 - cool luminous stars to hot faint stars only

C. large planets to small planets

D. near stars to distant stars

11. A white dwarf is supported against further collapse mainly by

A. ordinary gas pressure from hydrogen fusion

B. electron degeneracy pressure

C. ocean pressure

D. radiation from a companion planet

12. The observed redshift of many distant galaxies supports the idea that the Universe is

A. shrinking uniformly

B. static with Earth at the centre

C. expanding

D. inside the Solar System

13. Most weather occurs in the lowest atmospheric layer, the

A. stratosphere

B. mesosphere

C. troposphere

D. exosphere

14. Earth's magnetosphere reduces the direct impact of much of the

A. moonlight

B. ocean current

C. solar wind

D. rainfall

15. The large dark plains on the Moon are called

A. auroras

B. maria

C. continents

D. coronae

16. During a strong geomagnetic storm, which system is most likely to require special protection?

A. A paper map

B. Satellites and long power lines

C. A stone building with no electronics

D. A mechanical pencil

17. A geostationary satellite must orbit in the

A. polar plane with a 90-minute period

B. equatorial plane with any period

C. Moon's orbit

D. equatorial plane with a period equal to Earth's rotation

18. In a communication satellite, a transponder mainly

A. produces rocket fuel

B. receives, processes and retransmits signals

C. measures lunar craters only

D. creates gravity

19. Remote sensing means collecting information about a target

A. without direct physical contact

B. only by touching it

C. only from underground

D. without using radiation

20. The spacecraft subsystem that supplies electrical power commonly includes

A. solar arrays and batteries

B. heat shield and parachute only

C. rocket nozzle only

D. spectrograph only

21. Rocket thrust is produced when high-speed exhaust is expelled

A. in the same direction as gravity only

B. opposite to the direction of acceleration

C. without momentum

D. sideways in every direction equally

22. The main benefit of a multistage launch vehicle is that it can

A. avoid carrying propellant

B. discard empty structure and reduce mass during flight

C. stop after the first engine burn

D. remain inside the atmosphere

23. Escape velocity from a planet depends mainly on the planet's mass and

A. radius

B. colour

C. rotation direction only

D. number of moons only

24. Cryogenic rocket engines use propellants stored at

A. very low temperatures

B. room temperature only

C. temperatures above the Sun's surface

D. no controlled temperature

25. PSLV is especially associated with launching many satellites into

A. deep ocean trenches

B. the Sun's surface

C. polar and Sun-synchronous orbits

D. only geostationary transfer orbits

- 26.** Chandrayaan-1 is remembered for strong evidence of
- | | |
|---------------------------|--|
| A. liquid oceans on Venus | B. life on Mars |
| C. rings around Earth | D. water-related molecules on the Moon |
- 27.** Aditya-L1 observes the Sun from a halo orbit around the Sun-Earth
- | | |
|------------------------|-----------------------|
| A. L1 point | B. L4 point |
| C. Earth-Moon L2 point | D. geostationary ring |
- 28.** Gaganyaan is India's programme aimed at demonstrating
- | | |
|---------------------------------|---------------------------------|
| A. sample return from Mars | B. human spaceflight capability |
| C. a radio telescope on Jupiter | D. mining the asteroid belt |
- 29.** A planetary orbiter is best suited for
- | | |
|---|---|
| A. directly drilling every location | B. returning samples without any other system |
| C. repeated global or regional observations over time | D. removing a planet's atmosphere |
- 30.** A major purpose of a life-support system in human spaceflight is to control
- | | |
|---|----------------------|
| A. the Sun's mass | B. Earth's orbit |
| C. air, pressure, temperature and waste | D. stellar evolution |
- 31.** An asteroid sample-return mission is valuable because laboratory analysis on Earth can be
- | | |
|---|---------------------------------|
| A. performed without a sample | B. used to stop gravity |
| C. more detailed than most onboard analysis | D. done only with visible light |
- 32.** A practical early goal for future lunar exploration is to use local resources such as
- | | |
|-------------------------|--|
| A. dark matter for food | B. sunspots as landing pads |
| C. Earth's core as fuel | D. water ice for life support or propellant production |
- 33.** A satellite in circular motion has acceleration directed
- | | |
|---------------------------|--|
| A. along the tangent only | B. toward the centre of its orbit |
| C. away from the centre | D. nowhere because its speed is constant |
- 34.** For a wave, speed equals
- | | |
|------------------------------------|---------------------------------------|
| A. frequency divided by wavelength | B. amplitude multiplied by time |
| C. wavelength divided by energy | D. frequency multiplied by wavelength |
- 35.** A current-carrying wire produces
- | | |
|-------------------------------|-------------------------------|
| A. only a gravitational field | B. no field in any case |
| C. a sound wave in vacuum | D. a magnetic field around it |
- 36.** In vacuum, a spacecraft exchanges heat with its surroundings mainly by
- | | |
|---------------------------|--------------|
| A. conduction through air | B. radiation |
| C. convection through air | D. sound |
- 37.** In an experiment testing how lamp distance affects solar-cell output, the independent variable is
- | | |
|-------------------------------|-----------------------|
| A. measured electrical output | B. type of graph used |
| C. distance from the lamp | D. conclusion |
- 38.** A graph of temperature against time becomes less steep. This means the rate of temperature increase is
- | | |
|--------------------------|----------------------------|
| A. decreasing | B. increasing |
| C. constant and infinite | D. negative from the start |
- 39.** As of the 2026 information cut-off, Aditya-L1 was an operational mission studying the
- | | |
|---------|------------------------|
| A. Moon | B. Mars surface |
| C. Sun | D. Earth's oceans only |
- 40.** Space sustainability aims to preserve the space environment by reducing
- | | |
|--|---|
| A. the number of scientific observations | B. collision risk and long-lived debris |
| C. all satellite communication | D. Earth's gravity |

SECTION B - OLYMPIAD CHALLENGE (QUESTIONS 41-50)

These questions are intentionally difficult and may require calculations, graph reading or multi-step reasoning.

- 41.** A star has annual parallax 0.125 arcsecond. About how many light-years away is it? Use 1 parsec = 3.26 light-years.
- | | |
|---------------------|----------------------|
| A. 8.0 light-years | B. 0.41 light-years |
| C. 26.1 light-years | D. 260.8 light-years |
- 42.** A telescope aperture increases from 20 cm to 40 cm. At the same wavelength, its light-collecting area becomes
- | | |
|---------------------|----------------------|
| A. 2 times as large | B. 8 times as large |
| C. 4 times as large | D. 16 times as large |
- 43.** A planet orbits a Sun-like star with semi-major axis 4 AU. Using $T^2 = a^3$ with T in years, its period is
- | | |
|------------|-------------|
| A. 2 years | B. 16 years |
| C. 6 years | D. 8 years |
- 44.** A satellite moves from a circular orbit of radius R to one of radius 4R around the same planet. Its circular speed becomes
- | | |
|-----------------------------------|---------------|
| A. one-half of its original value | B. one-fourth |
| C. twice | D. four times |
- 45.** Star A and Star B have the same surface temperature, but Star A has four times the radius of Star B. The luminosity of A is approximately
- | | |
|------------|-----------------------|
| A. 4 times | B. 8 times |
| C. 2 times | D. 16 times that of B |
- 46.** A geostationary satellite sends a signal to a ground station. If the one-way path is about 36,000 km, the minimum one-way travel time is closest to
- | | |
|------------|----------|
| A. 0.012 s | B. 1.2 s |
| C. 0.12 s | D. 12 s |
- 47.** A rocket of total mass 5000 kg produces 75,000 N thrust. Ignoring drag and using $g = 10 \text{ m/s}^2$, its initial upward acceleration is
- | | |
|-----------------------|-----------------------|
| A. 15 m/s^2 | B. 5 m/s^2 |
| C. 10 m/s^2 | D. 25 m/s^2 |
- 48.** A convex lens of focal length 20 cm forms a real image of an object placed 60 cm away. The image distance is
- | | |
|----------|----------|
| A. 15 cm | B. 30 cm |
| C. 40 cm | D. 60 cm |
- 49.** A student measures orbital periods 1.00, 1.02, 0.99 and 1.01 s. The mean is
- | | |
|------------|------------|
| A. 1.005 s | B. 1.000 s |
| C. 1.020 s | D. 4.02 s |
- 50.** A solar storm reaches Earth shortly after a strong coronal mass ejection. Which chain is most scientifically reasonable?
- | | |
|--|--|
| A. Solar eruption -> Earth stops rotating -> tides vanish | B. Solar eruption -> disturbed magnetosphere -> possible satellite and radio effects |
| C. Solar eruption -> Moon changes orbit instantly -> eclipse | D. Solar eruption -> speed of light decreases |

MOCK PAPER 2

Senior Level (Classes IX-X) - 50 Questions - 50 Marks

Name		Roll Number	
School / Centre		Score	/ 50

Instructions: Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. Select the one best answer from A, B, C and D. Suggested time: 90 minutes.

SECTION A - CORE QUESTIONS

- An observer measures an object 35 degrees above the eastern horizon. The 35 degrees is its
 - azimuth
 - altitude
 - declination
 - right ascension
- A nearby star has annual parallax 0.20 arcsecond. Its distance is
 - 5 parsecs
 - 16.3 parsecs
 - 0.2 parsec
 - 50 parsecs
- Increasing a telescope aperture mainly improves its ability to
 - collect light and resolve fine detail
 - slow the speed of light
 - remove all atmospheric effects
 - magnify without limit
- Which radiation has a shorter wavelength than visible light but a longer wavelength than X-rays?
 - Radio waves
 - Ultraviolet
 - Microwaves
 - Infrared
- Why did the young Solar System become disk-shaped?
 - Gravity stopped acting
 - Rotation and collisions reduced motion perpendicular to the main plane
 - The Sun pushed all matter into a straight line
 - Planets formed before the Sun
- The greenhouse effect warms a planetary surface because some atmospheric gases
 - create nuclear fusion
 - absorb and re-emit outgoing infrared radiation
 - reflect all sunlight away
 - stop gravity
- According to Kepler's second law, a planet sweeps equal areas in equal times. Therefore it moves fastest near
 - perihelion
 - aphelion
 - the equinox
 - the celestial pole
- Most known asteroids orbit mainly between
 - Earth and the Moon
 - Uranus and Neptune
 - Mars and Jupiter
 - the Sun and Mercury
- A massive star can end its life in a core-collapse supernova and leave behind a
 - comet or asteroid
 - terrestrial planet only
 - red dwarf only
 - neutron star or black hole

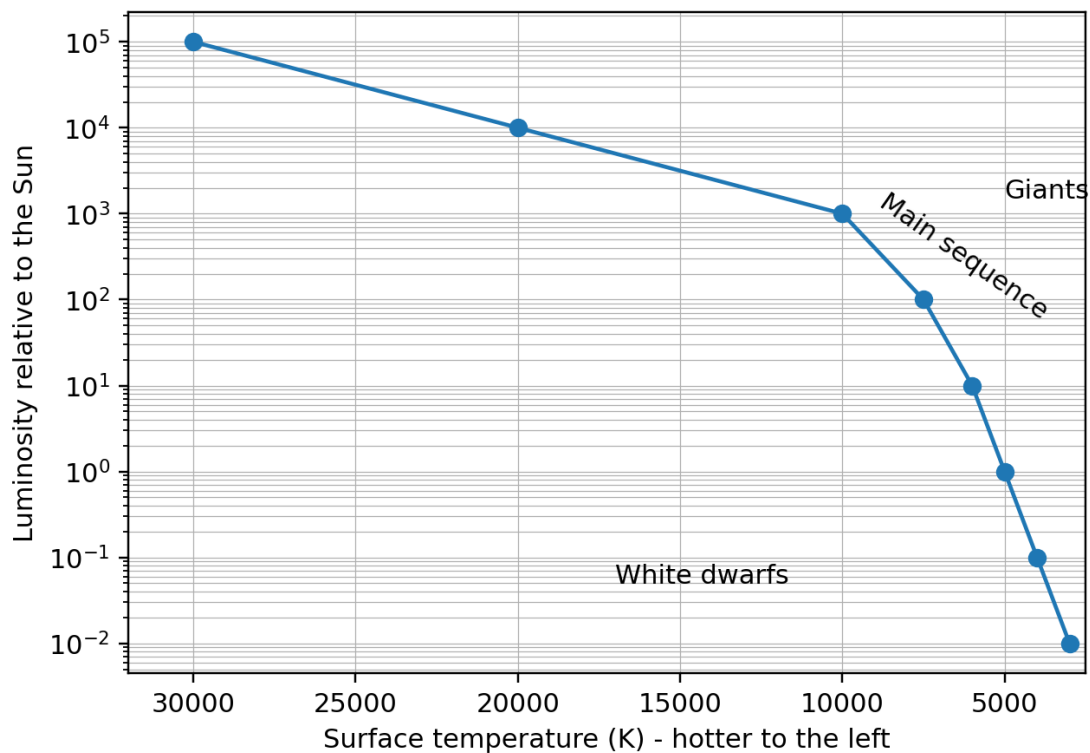


Figure 2. Simplified Hertzsprung-Russell diagram.

- 10.** Study Figure 2. The region labelled white dwarfs contains stars that are generally
- | | |
|--|------------------------------------|
| A. hot but faint | B. cool and very luminous |
| C. cool and faint because they are planets | D. hot and larger than supergiants |
- 11.** A neutron star has extremely high density because most of its matter is compressed mainly into
- | | |
|---------------------------|---------------------|
| A. neutrons | B. molecules of air |
| C. liquid hydrogen oceans | D. dust grains |
- 12.** The Big Bang model describes the early Universe as
- | | |
|---------------------------------------|---|
| A. hotter and denser than it is today | B. cold and empty forever |
| C. already containing mature galaxies | D. formed inside a black hole in the Milky Way as a proven fact |
- 13.** Earth's climate differs from day-to-day weather because climate describes
- | | |
|---------------------|-----------------------|
| A. only one storm | B. long-term patterns |
| C. the Moon's orbit | D. a telescope image |
- 14.** Auroras are produced when charged particles interact with gases mainly in Earth's
- | | |
|----------------|---|
| A. mantle | B. upper atmosphere near magnetic polar regions |
| C. ocean floor | D. inner core |
- 15.** The Moon preserves many ancient craters because it has
- | | |
|---|---------------------|
| A. very little atmosphere and surface erosion | B. strong rainfall |
| C. active oceans | D. dense vegetation |
- 16.** A sudden increase in high-energy solar radiation can cause short-term disturbances in
- | | |
|--------------------|--|
| A. Earth's mass | B. radio communication and satellite electronics |
| C. planetary order | D. the number of lunar maria |
- 17.** A polar-orbiting Earth observation satellite is useful because Earth rotates beneath it, allowing
- | | |
|---|--|
| A. the satellite to remain above one city | B. coverage of much of the globe over time |
| C. gravity to vanish | D. direct travel to Mars |
- 18.** A navigation receiver finds its position by using signals from
- | | |
|-----------------------|---|
| A. one telescope only | B. underwater cables only |
| C. the Moon's shadow | D. multiple satellites with accurately known timing |

- 19.** Vegetation can be distinguished in multispectral images because it reflects different amounts of radiation at
 - A. different times only
 - B. different gravitational constants
 - C. different wavelengths
 - D. different speeds of sound in vacuum
- 20.** Attitude control is used to
 - A. change the speed of light
 - B. eliminate orbital motion
 - C. orient a spacecraft and its instruments correctly
 - D. produce oxygen from vacuum automatically
- 21.** A rocket can operate in vacuum because it
 - A. pushes against atmospheric air
 - B. uses wings only
 - C. carries both fuel and oxidiser
 - D. needs ocean water
- 22.** A spent stage is separated mainly to
 - A. increase drag
 - B. reduce thrust to zero permanently
 - C. make the payload heavier
 - D. remove mass that is no longer useful
- 23.** Orbital insertion requires a spacecraft to gain the correct
 - A. temperature only
 - B. colour and mass only
 - C. radio name
 - D. speed and direction at the required altitude
- 24.** Liquid hydrogen and liquid oxygen are used in some cryogenic stages because they can provide
 - A. high specific impulse
 - B. zero exhaust velocity
 - C. no need for tanks
 - D. automatic reusability
- 25.** LVM3 launched which Indian lunar landing mission?
 - A. Chandrayaan-1
 - B. Chandrayaan-3
 - C. Mars Orbiter Mission
 - D. Aditya-L1
- 26.** Chandrayaan-3 demonstrated
 - A. human landing on Mars
 - B. sample return from Venus
 - C. a telescope at L1
 - D. soft landing and rover operations near the lunar south polar region
- 27.** XPoSat is designed to study the
 - A. ocean salinity only
 - B. Moon's surface with a rover
 - C. polarisation and spectra of cosmic X-ray sources
 - D. Sun using visible photographs only
- 28.** As of the 2026 information cut-off, Gaganyaan was best described as
 - A. under development and testing
 - B. a completed crewed lunar landing
 - C. a retired weather satellite
 - D. an operational Mars colony
- 29.** A lander is designed primarily to
 - A. remain permanently in solar orbit only
 - B. act as a launch pad on Earth
 - C. measure only distant stars
 - D. reach and operate on a planetary surface
- 30.** Astronauts in an orbiting space station appear weightless because they and the station are
 - A. beyond all gravity
 - B. made massless
 - C. in continuous free fall
 - D. not moving
- 31.** A kinetic impactor changes an asteroid's motion by
 - A. transferring momentum in a collision
 - B. removing its mass with sound
 - C. changing the speed of light
 - D. heating Earth's core
- 32.** A mission to the outer planets often requires radioisotope power because sunlight there is
 - A. stronger than near Earth
 - B. absent from the Universe
 - C. converted entirely to sound
 - D. much weaker
- 33.** If the net force on an object is zero, its momentum
 - A. must become zero
 - B. must increase
 - C. must reverse
 - D. remains constant

- 34.** A convex lens can form a real image on a screen when the object is
 A. exactly at the optical centre only
 B. always inside the focal length
 C. at infinity only
 D. beyond the focal point
- 35.** Electromagnetic induction occurs when changing magnetic flux produces
 A. mass
 B. temperature only
 C. a gravitational constant
 D. an emf
- 36.** The photoelectric effect shows that light can transfer energy in discrete packets called
 A. photons
 B. orbits
 C. molecules
 D. pressure waves
- 37.** A control variable should be kept constant so that
 A. all results become identical
 B. the effect of the independent variable can be isolated
 C. measurement is unnecessary
 D. the hypothesis is automatically true
- 38.** If two quantities are directly proportional, doubling one while other conditions remain fixed will
 A. double the other
 B. halve the other
 C. leave it unchanged
 D. make it zero
- 39.** In 2026, XPOSat continued to support research on
 A. ocean tides
 B. cosmic X-ray sources
 C. plate tectonics only
 D. GPS timing
- 40.** Commercial spaceflight refers to space activities in which
 A. only governments may participate
 B. no safety rules apply
 C. private companies provide vehicles or services
 D. scientific goals are forbidden

SECTION B - OLYMPIAD CHALLENGE (QUESTIONS 41-50)

These questions are intentionally difficult and may require calculations, graph reading or multi-step reasoning.

- 41.** A star has annual parallax 0.200 arcsecond. About how many light-years away is it? Use 1 parsec = 3.26 light-years.
 A. 5.0 light-years
 B. 0.65 light-years
 C. 163.0 light-years
 D. 16.3 light-years
- 42.** Two telescopes observe at the same wavelength. Telescope A has twice the aperture of telescope B. The diffraction-limited angle of A is
 A. twice that of B
 B. four times that of B
 C. the same
 D. half that of B
- 43.** A planet orbits a Sun-like star with semi-major axis 9 AU. Using $T^2 = a^3$ with T in years, its period is
 A. 3 years
 B. 81 years
 C. 27 years
 D. 11 years
- 44.** A planet has four times Earth's mass and the same radius. Its escape velocity is
 A. four times Earth's
 B. half Earth's
 C. twice Earth's
 D. the same
- 45.** Study Figure 2. A star moves from the main sequence to the red-giant region. Its surface becomes cooler but luminosity increases mainly because its
 A. mass becomes zero
 B. radius increases greatly
 C. distance from Earth decreases
 D. parallax vanishes
- 46.** A satellite image has 20 m spatial resolution. A square field 100 m on each side covers approximately
 A. 5 pixels
 B. 10 pixels
 C. 25 pixels
 D. 100 pixels
- 47.** A launch vehicle has mass 2.0×10^5 kg and thrust 3.0×10^6 N. With $g = 10 \text{ m/s}^2$, its initial acceleration is
 A. 5 m/s^2
 B. 15 m/s^2
 C. 10 m/s^2
 D. 1.5 m/s^2

- 48.** A radio signal has frequency 150 MHz. Its wavelength in vacuum is
- A. 0.5 m
 - B. 20 m
 - C. 2 m
 - D. 200 m
- 49.** An instrument gives readings 12.1, 12.2, 12.1, 12.2 cm when the accepted value is 10.0 cm. The readings are
- A. accurate but imprecise
 - B. precise but inaccurate
 - C. both accurate and precise
 - D. neither repeatable nor biased
- 50.** XPoSat measures X-ray polarisation while Aditya-L1 observes the Sun. Which statement correctly distinguishes them?
- A. Both are lunar rovers
 - B. Both are navigation satellites
 - C. XPoSat studies cosmic X-ray sources; Aditya-L1 studies solar phenomena
 - D. Aditya-L1 studies only Earth weather

MOCK PAPER 3

Senior Level (Classes IX-X) - 50 Questions - 50 Marks

Name		Roll Number	
School / Centre		Score	/ 50

Instructions: Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. Select the one best answer from A, B, C and D. Suggested time: 90 minutes.

SECTION A - CORE QUESTIONS

- Right ascension on the celestial sphere is most closely analogous to which Earth coordinate?
 - longitude
 - latitude
 - altitude
 - depth
- A nearby star has annual parallax 0.10 arcsecond. Its distance is
 - 32.6 parsecs
 - 0.1 parsec
 - 100 parsecs
 - 10 parsecs
- A radio telescope commonly uses a large dish because radio waves are
 - sound waves in air
 - focused by the dish onto a receiver
 - visible only at night
 - blocked by every cloud
- An astronomer wants to study cool interstellar dust hidden by visible clouds. A useful band is often
 - infrared
 - sound
 - ultrasound
 - green light only
- The inner planets became mainly rocky because near the young Sun
 - hydrogen froze first
 - gravity was absent
 - all metals evaporated forever
 - only heat-resistant materials could readily condense
- A thick carbon-dioxide atmosphere contributes strongly to the high surface temperature of
 - Mercury
 - Venus
 - Mars only because it is closest
 - the Moon
- If a planet is farther from the Sun, Kepler's third law predicts that its orbital period is generally
 - shorter
 - zero
 - longer
 - independent of distance
- An exoplanet is a planet that
 - orbits Earth
 - orbits a star other than the Sun
 - has left the Galaxy
 - is smaller than every asteroid
- A red giant forms when a Sun-like star
 - exhausts much of the hydrogen in its core and its outer layers expand
 - first begins to form from a nebula
 - becomes a planet
 - loses all gravity
- A red giant is luminous even though its surface is relatively cool mainly because it has a very large
 - parallax
 - magnetic pole
 - surface area
 - orbital speed
- The event horizon of a black hole is the boundary beyond which
 - gravity becomes zero
 - escape would require faster-than-light motion
 - all time in the Universe stops everywhere
 - matter loses mass
- The cosmic microwave background is interpreted as
 - sunlight reflected by planets
 - relic radiation from the early Universe

C. radio noise produced only by Earth

D. light from one nearby star

13. The ozone layer that absorbs much ultraviolet radiation is concentrated mainly in the

- A. troposphere only at sea level
C. outer core

- B. stratosphere
D. mantle

14. A coronal mass ejection directed at Earth can disturb

- A. the mass of the Sun
C. Earth's orbital period

- B. the Moon's crater count
D. satellite operations and power systems

15. The leading model for Moon formation is a

- A. giant impact between early Earth and a large body
C. direct split caused by sunlight

- B. capture of a modern comet
D. collision between two galaxies

16. Earth-observing instruments detect changes in vegetation most directly by measuring

- A. sound from leaves
C. gravity from each tree

- B. reflected radiation in selected wavelength bands
D. the age of the Sun

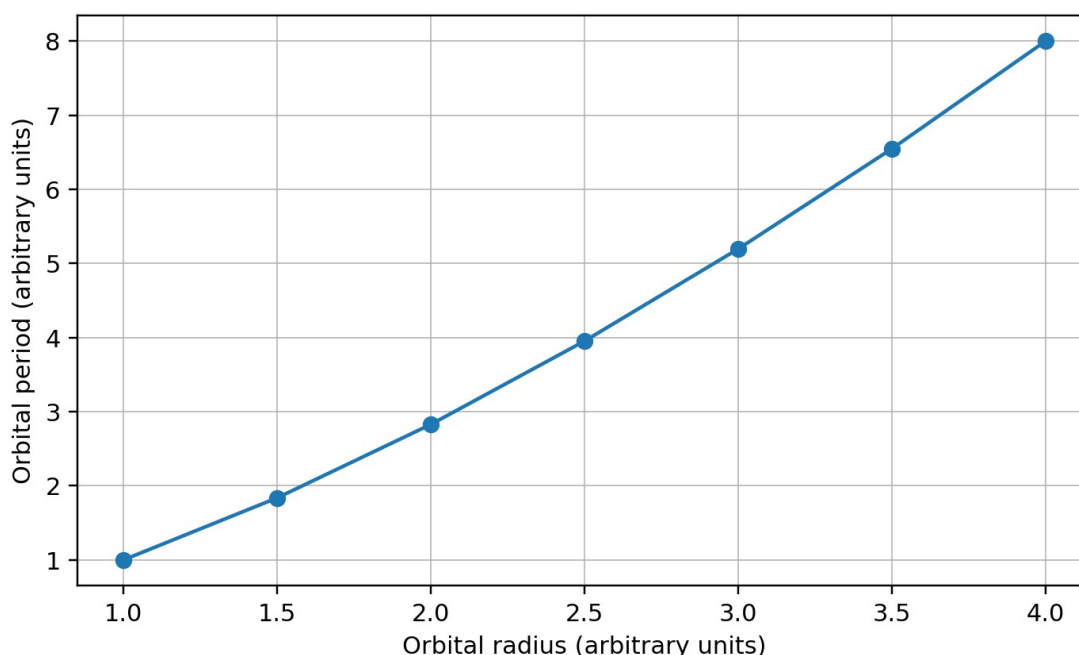


Figure 3. General relation between orbital radius and orbital period.

17. Study Figure 3. As orbital radius increases, the orbital period

- A. decreases to zero
C. remains constant

- B. increases
D. first decreases and then vanishes

18. A major advantage of geostationary communication satellites is that ground antennas can

- A. operate without power
C. see the satellite from both poles equally well

- B. ignore signal delay completely
D. point in a nearly fixed direction

19. A passive remote-sensing instrument mainly measures

- A. a signal it always transmits first
C. the spacecraft mass

- B. only sound echoes
D. naturally emitted or reflected radiation

20. A thermal-control system is needed because spacecraft

- A. must keep equipment within safe temperature limits
C. always remain at room temperature

- B. are never heated by sunlight
D. do not radiate energy

21. Specific impulse is a measure related to a rocket engine's

- A. paint colour
C. radio frequency only

- B. launch-pad height
D. propellant efficiency

22. In a typical launch, the upper stage is used after lower stages to provide

- A. additional velocity and accurate orbital injection
C. weather forecasting only
- B. ocean navigation
D. a permanent launch tower
- 23.** Compared with circular orbital speed at the same radius, escape speed is
A. $\sqrt{2}$ times as large
C. equal in every case
- B. half as large
D. twice as large exactly
- 24.** A reusable launch vehicle is designed to
A. recover and fly major hardware again
C. remain permanently on the launch pad
- B. use no propulsion
D. avoid all heating during re-entry
- 25.** GSLV differs from PSLV partly because GSLV versions are designed to place heavier payloads toward
A. submarine orbit
C. only low-altitude aircraft routes
- B. the asteroid belt directly
D. geosynchronous transfer orbit
- 26.** The Mars Orbiter Mission studied Mars from
A. the surface using a rover
C. orbit
- B. inside its atmosphere with a balloon only
D. a station at the Sun
- 27.** A key advantage of the Sun-Earth L1 region for Aditya-L1 is
A. nearly continuous viewing of the Sun
C. permanent darkness
- B. zero gravity everywhere
D. no communication delay
- 28.** Chandrayaan-4 is planned primarily to demonstrate
A. a solar observatory at L1
C. X-ray polarimetry
- B. lunar sample return technologies
D. an Earth weather constellation only
- 29.** A rover extends exploration by
A. changing a planet's orbit
C. moving between different surface locations
- B. creating artificial gravity for Earth
D. replacing all orbiters
- 30.** Exercise is important on long space missions mainly to reduce loss of
A. radio signals
C. solar radiation
- B. orbital energy
D. muscle and bone strength
- 31.** Space telescopes can observe wavelengths blocked by Earth's atmosphere, such as much of the
A. audible sound band
C. seismic band only
- B. ocean-wave band
D. X-ray and ultraviolet bands
- 32.** Autonomous navigation is especially useful for distant probes because
A. radio travels infinitely fast
C. communication delays prevent instant control
- B. gravity stops beyond Mars
D. computers cannot operate on Earth
- 33.** Gravitational force between two masses varies inversely as the
A. first power of separation
C. sum of their radii only
- B. cube of separation
D. square of their separation
- 34.** Diffraction becomes more noticeable when an opening is
A. comparable in size to the wavelength
C. perfectly opaque
- B. millions of times wider only
D. at absolute zero
- 35.** Two like electric charges
A. attract
C. have no interaction
- B. repel
D. always become neutral
- 36.** A hotter blackbody emits its peak radiation at
A. longer wavelength only
C. shorter wavelength
- B. the same wavelength
D. zero frequency
- 37.** Repeating a measurement mainly helps to
A. estimate random variation and improve reliability
- B. remove every systematic error automatically

- C. change the physical law
D. avoid recording data
- 38.** Study Figure 4. Between 0 and 6 minutes, the surface temperature rises most rapidly during
A. the last two minutes
B. the first two minutes
C. only at 6 minutes
D. at a constant rate throughout
- 39.** The 2026 Gaganyaan parachute tests were intended to help qualify systems for safe
A. solar imaging
B. lunar drilling
C. crew-module descent and splashdown
D. asteroid mining
- 40.** A space policy can regulate issues such as licensing, safety and
A. responsible use of orbital resources
B. the laws of motion
C. the speed of light
D. the existence of gravity

SECTION B - OLYMPIAD CHALLENGE (QUESTIONS 41-50)

These questions are intentionally difficult and may require calculations, graph reading or multi-step reasoning.

- 41.** A star has annual parallax 0.050 arcsecond. About how many light-years away is it? Use 1 parsec = 3.26 light-years.
A. 20.0 light-years
B. 0.16 light-years
C. 652.0 light-years
D. 65.2 light-years
- 42.** A 25 cm telescope collects 900 photons from a source in 1 minute. Under identical conditions, a 50 cm telescope should collect approximately
A. 1800 photons
B. 450 photons
C. 3600 photons
D. 7200 photons
- 43.** A planet orbits a Sun-like star with semi-major axis 16 AU. Using $T^2 = a^3$ with T in years, its period is
A. 4 years
B. 256 years
C. 18 years
D. 64 years
- 44.** A planet has Earth's mass but one-fourth Earth's radius. Its surface escape velocity is
A. four times Earth's
B. half Earth's
C. twice Earth's
D. the same
- 45.** Two stars have the same radius. Star X has twice the surface temperature of Star Y. Treating them as blackbodies, X has approximately
A. 2 times
B. 4 times
C. 16 times the luminosity
D. 8 times
- 46.** A satellite completes 15 orbits in 24 hours. Its average orbital period is
A. 90 minutes
B. 96 minutes
C. 100 minutes
D. 120 minutes
- 47.** A rocket expels 20 kg of gas each second at 2500 m/s relative to the rocket. Ignoring pressure thrust, the thrust is
A. 5,000 N
B. 50,000 N
C. 125,000 N
D. 2,520 N
- 48.** A charged particle moves perpendicular to a uniform magnetic field. If its speed doubles while charge and field remain fixed, the magnetic force becomes
A. four times as large
B. half as large
C. twice as large
D. unchanged
- 49.** A graph of T^2 against r^3 for circular satellite orbits is a straight line through the origin. This supports the relation
A. T^2 is proportional to r^3
B. T is proportional to r^3
C. T^3 is proportional to r^2
D. T^2 is proportional to $1/r^3$
- 50.** A future lunar sample-return mission requires landing, sampling, ascent, rendezvous and Earth return. Which added capability is most essential compared with a one-way lander?
A. Only taking photographs
B. Only measuring temperature
C. Launching material from the Moon and returning it safely to Earth
D. Only receiving radio signals

MOCK PAPER 4

Senior Level (Classes IX-X) - 50 Questions - 50 Marks

Name		Roll Number	
School / Centre		Score	/ 50

Instructions: Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. Select the one best answer from A, B, C and D. Suggested time: 90 minutes.

SECTION A - CORE QUESTIONS

- The point on the sky directly above an observer is called the
 - nadir
 - zenith
 - ecliptic
 - celestial equator
- A nearby star has annual parallax 0.50 arcsecond. Its distance is
 - 2 parsecs
 - 0.5 parsec
 - 6.5 parsecs
 - 20 parsecs
- A space telescope can obtain sharper images than a similar ground telescope mainly because it is above
 - Earth's gravity
 - the electromagnetic spectrum
 - atmospheric turbulence
 - all cosmic radiation
- The wavelength of electromagnetic radiation increases in the order
 - visible, ultraviolet, X-ray, gamma ray
 - infrared, microwave, radio, gamma ray
 - radio, microwave, infrared, visible
 - gamma ray, X-ray, ultraviolet, visible
- Evidence that planets formed from a common rotating disk includes their mostly
 - equal sizes
 - perfectly circular paths
 - identical atmospheres
 - same-direction, nearly coplanar orbits
- Large shield volcanoes on Mars indicate that the planet
 - had major volcanic activity in its past
 - is a gas giant
 - never contained internal heat
 - has no crust
- For an elliptical orbit, the point farthest from the Sun is called
 - perihelion
 - zenith
 - perigee
 - aphelion
- A comet develops a coma and tail when it approaches the Sun because
 - ices heat and release gas and dust
 - it enters Earth's atmosphere
 - its rock melts into lava only
 - the Sun stops rotating
- A planetary nebula is produced when a low- or medium-mass star
 - collides with every planet
 - turns into a galaxy
 - ejects its outer layers near the end of its life
 - creates a new Big Bang
- Two main-sequence stars are compared. The hotter blue star is generally
 - less luminous and always older
 - a planet
 - more luminous and more massive
 - a white dwarf by definition
- A pulsar is best described as a rapidly rotating
 - comet with a tail
 - galaxy cluster
 - young planet
 - magnetised neutron star producing regular beams
- Hubble's law states that, on large scales, a more distant galaxy generally has a greater
 - recession speed
 - rotation of Earth
 - surface gravity
 - apparent angular size

- 13.** Earth's internal magnetic field is generated mainly by motion of conducting material in the
A. solid crust
B. liquid outer core
C. upper atmosphere
D. oceans
- 14.** Space weather is driven mainly by changing conditions originating from the
A. Earth's oceans
B. Milky Way centre only
C. Moon's maria
D. Sun
- 15.** A lunar eclipse occurs when
A. Earth is between the Sun and Moon
B. Moon is between Sun and Earth
C. Mars blocks the Moon
D. Sun is between Earth and Moon
- 16.** An image taken repeatedly over the same region is especially useful for studying
A. change over time
B. only the original launch date
C. the speed of light
D. a star's internal pressure
- 17.** A satellite in low Earth orbit usually completes an orbit in roughly
A. one month
B. about 90 to 120 minutes
C. one year
D. 24 days
- 18.** Satellite communication at very high frequencies can be weakened by
A. rain absorption and scattering
B. the absence of sound in space only
C. Earth's rotation stopping
D. lunar tides
- 19.** Radar is an active remote-sensing system because it
A. sends microwave pulses and records the return
B. uses sunlight only
C. requires no energy
D. detects only visible colour
- 20.** Telemetry is information sent from a spacecraft to the ground about
A. only the weather at launch site
B. the mass of Earth
C. future events
D. its status and measurements
- 21.** The combustion chamber of a chemical rocket converts chemical energy mainly into
A. gravitational shielding
B. visible mass loss only
C. nuclear fusion in all engines
D. thermal energy and high-speed exhaust motion
- 22.** Stage separation must be carefully timed to avoid
A. collision between stages and loss of control
B. the electromagnetic spectrum
C. Earth's seasons
D. star formation
- 23.** A spacecraft can remain in orbit while continuously falling because Earth's surface
A. stops gravity
B. curves away beneath its path
C. moves faster than light
D. becomes flat
- 24.** One major challenge of reusable vehicles is
A. the absence of gravity in orbit
B. that fuel has no mass
C. surviving re-entry heating and landing accurately
D. that radio waves cannot travel in space
- 25.** India's heaviest operational launch vehicle in the syllabus is
A. LVM3
B. SLV-3
C. ASLV
D. Sounding rocket only
- 26.** The Chandrayaan programme is focused mainly on exploration of the
A. Sun
B. Moon
C. Jupiter
D. asteroid belt only
- 27.** XPoSat observations are especially relevant to compact sources such as
A. continents and oceans
B. comet tails only
C. black holes and neutron stars
D. weather clouds

- 28.** India's approved Venus Orbiter Mission is intended to study Venus, including its
- | | |
|---------------------------|-------------------------|
| A. rings and moons only | B. human settlements |
| C. surface and atmosphere | D. ocean tides on Earth |
- 29.** A flyby mission gains data while
- | | |
|--|--|
| A. landing and remaining permanently | B. destroying the target |
| C. passing a target without entering orbit | D. stopping relative motion completely |
- 30.** A space station in low Earth orbit requires periodic reboots mainly because thin atmospheric drag
- | | |
|-------------------------------|-------------------------------------|
| A. gradually lowers its orbit | B. makes gravity zero |
| C. increases its mass | D. reverses its direction instantly |
- 31.** A deep-space mission often uses a gravity assist to
- | | |
|---------------------|---|
| A. turn off gravity | B. change speed and direction using a planet's motion and gravity |
| C. avoid navigation | D. stop the planet |
- 32.** A telescope placed at a stable space location can benefit from
- | | |
|---|------------------------------------|
| A. reduced atmospheric interference and long observations | B. stronger atmospheric turbulence |
| C. unlimited aperture without engineering | D. zero need for thermal control |
- 33.** Work done by a constant force parallel to displacement is
- | | |
|--------------------------------|-------------------------------------|
| A. force divided by time | B. force multiplied by displacement |
| C. mass multiplied by velocity | D. pressure multiplied by area only |
- 34.** A telescope objective with a larger diameter has a smaller diffraction-limited angle because resolution varies approximately as
- | | |
|-----------------------------------|--------------------------------------|
| A. aperture divided by wavelength | B. wavelength multiplied by aperture |
| C. aperture squared only | D. wavelength divided by aperture |
- 35.** A charged particle moving perpendicular to a magnetic field experiences a force
- | | |
|------------------------------|---|
| A. parallel to velocity only | B. perpendicular to both its velocity and the field |
| C. always zero | D. toward Earth only |
- 36.** Radioactive decay is described as a process in which an unstable nucleus
- | | |
|--|---------------------|
| A. gains unlimited mass | B. stops all motion |
| C. changes spontaneously and emits radiation | D. becomes a planet |
- 37.** If a measuring instrument always reads 2 units too high, the error is mainly
- | | |
|-----------------|----------------------|
| A. random only | B. a unit conversion |
| C. a hypothesis | D. systematic |

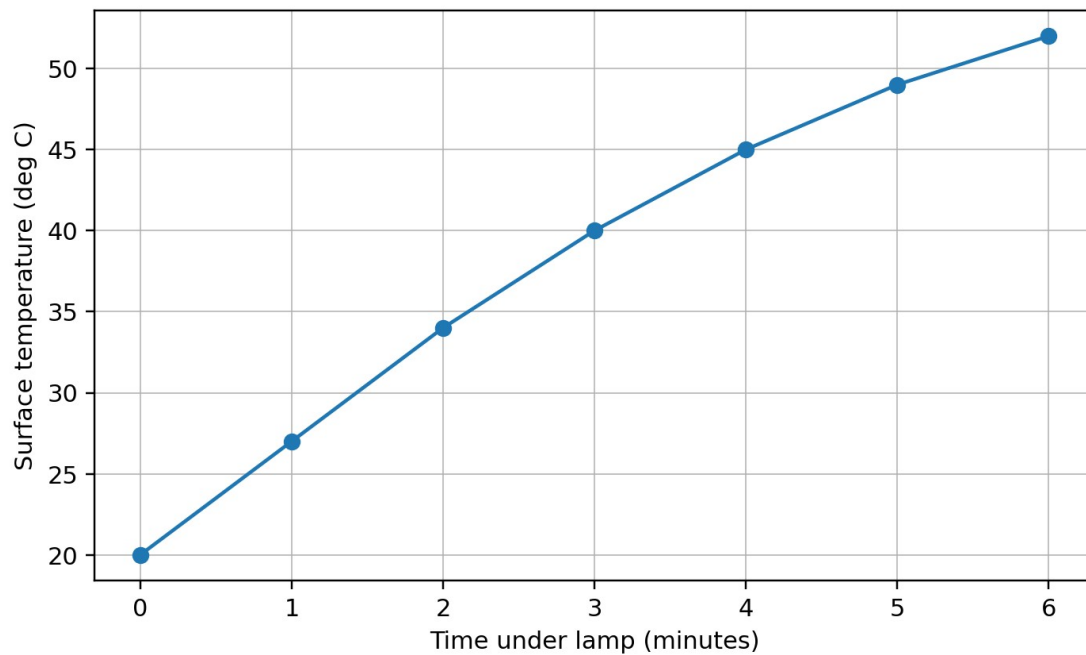


Figure 4. Temperature of a model surface heated by a lamp.

- 38.** A line of best fit is used to
- A. force every point onto one value
 - B. remove units
 - C. show the overall trend in scattered data
 - D. prove causation automatically
- 39.** NASA's Artemis II mission in 2026 carried astronauts on a flight
- A. to the surface of Mars
 - B. to Venus orbit
 - C. outside the Milky Way
 - D. around the Moon
- 40.** Emerging satellite technologies increasingly use onboard automation to
- A. eliminate physical limits
 - B. process data and support faster decisions
 - C. avoid all ground communication
 - D. stop orbital decay entirely

SECTION B - OLYMPIAD CHALLENGE (QUESTIONS 41-50)

These questions are intentionally difficult and may require calculations, graph reading or multi-step reasoning.

- 41.** A star has annual parallax 0.400 arcsecond. About how many light-years away is it? Use 1 parsec = 3.26 light-years.
- A. 8.1 light-years
 - B. 2.5 light-years
 - C. 1.30 light-years
 - D. 81.5 light-years
- 42.** A telescope resolves two stars separated by 1.2 arcsecond. A second telescope of the same design has three times the aperture and observes at the same wavelength. Its approximate resolution limit is
- A. 3.6 arcseconds
 - B. 0.4 arcsecond
 - C. 1.2 arcseconds
 - D. 0.13 arcsecond
- 43.** A planet orbits a Sun-like star with semi-major axis 25 AU. Using $T^2 = a^3$ with T in years, its period is
- A. 5 years
 - B. 625 years
 - C. 125 years
 - D. 27 years
- 44.** A satellite has period T in a circular orbit of radius r. At radius 9r, its period will be
- A. 9T
 - B. 3T
 - C. 27T
 - D. 81T
- 45.** A spectral line of laboratory wavelength 500 nm is observed at 501 nm. Using v/c about $\Delta\lambda/\lambda$, the recession speed is closest to
- A. 60 km/s
 - B. 600 km/s
 - C. 3000 km/s
 - D. 6 km/s
- 46.** A navigation signal arrives 0.070 s after transmission. Using $c = 3.0 \times 10^8$ m/s, the signal path length is
- A. 2,100 km
 - B. 210,000 km

C. 21,000 km

D. 4,286 km

47. A stage has dry mass 2 tonnes and carries 8 tonnes of propellant. After all propellant is used, the mass ratio initial/final is

A. 4

B. 8

C. 10

D. 5

48. A black surface absorbs 600 W and radiates 450 W. Its rate of increase of internal energy is

A. 1050 W

B. -150 W

C. 75 W

D. 150 W

49. Study Figure 4. The average temperature rise from 0 to 4 minutes is

A. 8 deg C per minute

B. 4 deg C per minute

C. 6.25 deg C per minute

D. 11.25 deg C per minute

50. A commercial satellite operator plans a large constellation. Which action best supports space sustainability?

A. Ignore inactive satellites

B. Place all satellites in exactly one orbit with no coordination

C. Provide collision avoidance, tracking and end-of-life disposal plans

D. Disable communication after launch

MOCK PAPER 5

Senior Level (Classes IX-X) - 50 Questions - 50 Marks

Name		Roll Number	
School / Centre		Score	/ 50

Instructions: Questions 1-40 form the Core Section. Questions 41-50 form the Olympiad Challenge. Select the one best answer from A, B, C and D. Suggested time: 90 minutes.

SECTION A - CORE QUESTIONS

- A star crossing the local north-south line in the sky is said to cross the
 - meridian
 - horizon
 - ecliptic
 - equator of Earth
- A nearby star has annual parallax 0.05 arcsecond. Its distance is
 - 20 parsecs
 - 65.2 parsecs
 - 0.05 parsec
 - 200 parsecs
- The Rayleigh criterion is connected most directly with a telescope's
 - fuel efficiency
 - angular resolution
 - orbital period
 - colour index
- A spectrograph is used primarily to measure
 - how intensity varies with wavelength
 - only the total mass of a star
 - the sound from a planet
 - the age of a telescope
- The Sun formed at the centre of the solar nebula mainly because
 - magnetism removed its mass
 - comets pushed it inward
 - most mass collected there under gravity
 - the planets existed first
- The giant planets have lower average density than terrestrial planets mainly because they contain much more
 - hydrogen and helium
 - iron and nickel
 - granite
 - liquid silicate rock only
- A planet in a circular orbit remains in orbit because gravity provides the required
 - buoyant force
 - frictional force
 - electrostatic shielding
 - centripetal force
- A small Solar System body whose orbit brings it close to Earth is classified as a
 - white dwarf
 - geostationary satellite
 - solar flare
 - Near-Earth Object
- The initial mass of a star is especially important because it strongly controls its
 - lifetime and final remnant
 - right ascension
 - distance from Earth only
 - constellation name
- Which pair of quantities is plotted on a basic H-R diagram?
 - Luminosity and surface temperature
 - Mass and distance from Earth
 - Right ascension and declination
 - Orbital period and eccentricity
- Which stellar remnant has the smallest typical radius?
 - Black hole event horizon for a stellar-mass remnant
 - red giant
 - white dwarf
 - main-sequence Sun-like star
- Dark matter is inferred mainly because visible matter alone cannot explain observed
 - sunrise times
 - gravitational effects in galaxies and clusters
 - colours of rainbows
 - phases of the Moon

- 13.** Earth observation satellites help monitor climate by repeatedly measuring
 A. only star positions
 B. only underground magma directly
 C. clouds, oceans, land and atmospheric properties
 D. the mass of the Moon
- 14.** The magnetosphere is compressed more strongly on the side facing the
 A. Moon
 B. north celestial pole
 C. Milky Way centre
 D. Sun
- 15.** The same side of the Moon generally faces Earth because the Moon's rotation period is
 A. zero
 B. one year
 C. half its orbital period
 D. equal to its orbital period around Earth
- 16.** Which observation best indicates a space-weather effect rather than ordinary weather?
 A. A local thunderstorm
 B. A geomagnetic disturbance following a solar eruption
 C. A monsoon rain band
 D. A sea breeze
- 17.** For two circular Earth orbits, the satellite in the lower orbit generally moves
 A. slower
 B. at the same speed always
 C. backwards
 D. faster
- 18.** A navigation satellite must carry a highly accurate
 A. chemical battery only
 B. clock
 C. camera lens only
 D. parachute
- 19.** Spatial resolution describes the
 A. number of satellites in orbit
 B. smallest ground detail that can be distinguished
 C. time since launch
 D. amount of propellant
- 20.** The command subsystem allows ground controllers to
 A. increase the Sun's brightness
 B. move Earth's axis
 C. change physical constants
 D. send instructions to the spacecraft
- 21.** A rocket nozzle helps produce thrust by
 A. stopping the exhaust
 B. increasing payload mass
 C. reflecting sunlight
 D. accelerating expanding gases
- 22.** A payload fairing is usually discarded after the rocket leaves the dense atmosphere because
 A. the payload becomes heavier
 B. gravity disappears
 C. the orbit is complete immediately
 D. aerodynamic protection is no longer required
- 23.** If a planet has the same mass as Earth but twice Earth's radius, its escape velocity is
 A. twice as large
 B. unchanged
 C. four times as large
 D. smaller by a factor of $\sqrt{2}$
- 24.** Cryogenic tanks need insulation mainly to reduce
 A. gravity
 B. radio signals
 C. orbital period
 D. heat entering and boiling the propellants
- 25.** A launch vehicle is selected mainly according to the payload mass and required
 A. paint colour
 B. school grade
 C. orbit
 D. constellation name
- 26.** A major engineering objective of Chandrayaan-3 was to demonstrate
 A. crewed lunar return
 B. construction of a space station
 C. safe landing and roving on the lunar surface
 D. X-ray polarimetry
- 27.** Aditya-L1 carries instruments to study the solar atmosphere, solar wind and
 A. Earth's oceans only
 B. magnetic and energetic phenomena
 C. lunar soil only
 D. galaxy rotation only

- 28.** India's future space roadmap includes development toward a national space station called the
- | | |
|----------------------------------|--------------------------|
| A. International Lunar Telescope | B. Aryabhata Sea Station |
| C. Bharatiya Antariksh Station | D. Mars Polar Base |
- 29.** Deep-space communication becomes difficult mainly because signals
- | | |
|---|--|
| A. cannot cross vacuum | B. move slower than sound |
| C. become weak and experience long travel times | D. are blocked by all planets at all times |
- 30.** Radiation protection becomes more difficult beyond low Earth orbit because astronauts leave much of Earth's
- | | |
|---------------------|-----------------------------|
| A. ocean pressure | B. tropospheric rain |
| C. surface friction | D. magnetospheric shielding |
- 31.** Planetary defence includes
- | | |
|--|--|
| A. detecting and characterising potentially hazardous near-Earth objects | B. moving all asteroids into Earth orbit |
| C. blocking every meteor with satellites | D. stopping the Solar System |
- 32.** Future Mars missions must manage dust because it can affect
- | | |
|---|-----------------------|
| A. the value of pi | B. Earth's day length |
| C. solar panels, mechanisms and instruments | D. the speed of light |
- 33.** Momentum is conserved in an isolated system because the total external
- | | |
|-----------------------|-------------------------|
| A. mass is zero | B. temperature is fixed |
| C. volume is infinite | D. force is zero |
- 34.** Refraction occurs because light changes speed when it enters a medium with a different
- | | |
|---------------------------------------|---------------------|
| A. mass density only and nothing else | B. refractive index |
| C. temperature only | D. magnetic pole |
- 35.** A spacecraft magnetometer is designed to measure
- | | |
|------------------------|-------------------|
| A. air pressure only | B. magnetic field |
| C. visible colour only | D. mass flow only |
- 36.** Temperature is related to the average kinetic energy of particles in a substance. Increasing temperature generally means
- | | |
|--|---------------------------------------|
| A. zero particle motion | B. less internal energy in every case |
| C. greater average random kinetic energy | D. no radiation |
- 37.** The best scientific conclusion is one that
- | | |
|--|--|
| A. ignores conflicting data | B. states more precision than measured |
| C. is supported by the collected evidence and acknowledges uncertainty | D. cannot be tested |
- 38.** A result far from the pattern of the other measurements is called an
- | | |
|-------------|------------------|
| A. axis | B. outlier |
| C. constant | D. control group |
- 39.** A major concern in the growing commercial space sector is responsible management of
- | | |
|-------------------------------|-----------------------|
| A. Earth's continents | B. star colours |
| C. orbital traffic and debris | D. the speed of light |
- 40.** End-of-life disposal planning is important because inactive satellites can become
- | | |
|-------------------|-----------------|
| A. orbital debris | B. new stars |
| C. natural moons | D. solar flares |

SECTION B - OLYMPIAD CHALLENGE (QUESTIONS 41-50)

These questions are intentionally difficult and may require calculations, graph reading or multi-step reasoning.

- 41.** A star has annual parallax 0.080 arcsecond. About how many light-years away is it? Use 1 parsec = 3.26 light-years.
- | | |
|---------------------|----------------------|
| A. 12.5 light-years | B. 40.8 light-years |
| C. 0.26 light-years | D. 407.5 light-years |

42. A star is observed through a filter that transmits 60% of incident light. If 5000 photons reach the filter, the ideal detector receives

- A. 3000 photons
B. 2000 photons
C. 500 photons
D. 8333 photons

43. A planet orbits a Sun-like star with semi-major axis 36 AU. Using $T^2 = a^3$ with T in years, its period is

- A. 6 years
B. 216 years
C. 1296 years
D. 38 years

44. At a given orbital radius, escape speed is 11.2 km/s. The circular orbital speed there is closest to

- A. 5.6 km/s
B. 11.2 km/s
C. 7.9 km/s
D. 15.8 km/s

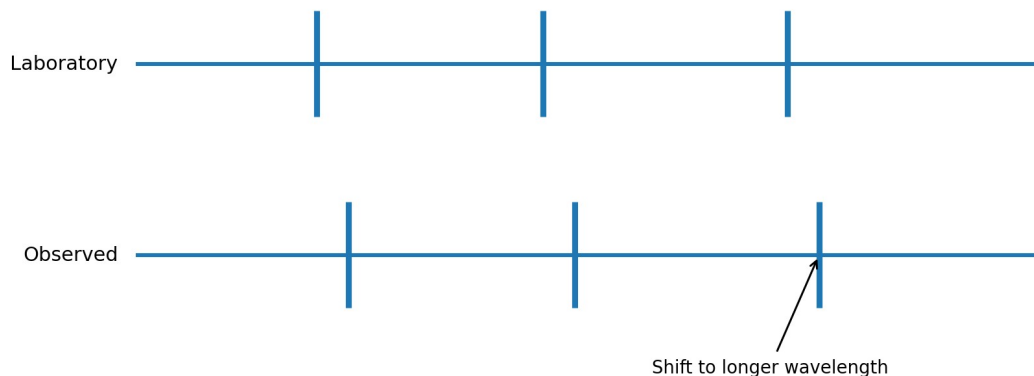


Figure 5. Comparison of laboratory and observed spectral lines.

45. Study Figure 5. The observed spectral lines are shifted to the right relative to laboratory lines. This most directly indicates

- A. blueshift and approach
B. redshift and recession
C. no relative motion
D. only a larger telescope aperture

46. A detector records reflected intensities 80 units in near-infrared and 20 units in red. The normalised difference (NIR-red)/(NIR+red) is

- A. 0.60
B. 0.75
C. 0.25
D. 1.00

47. A reusable booster descends from 1000 m with speed 50 m/s. If it decelerates uniformly to rest, the minimum deceleration magnitude is

- A. 2.5 m/s²
B. 0.8 m/s²
C. 1.25 m/s²
D. 5 m/s²

48. A photon has wavelength 500 nm. A photon of wavelength 250 nm has

- A. twice the energy
B. half the energy
C. four times the energy
D. the same energy

49. Five measurements are 18, 19, 19, 20 and 54. The median is

- A. 20
B. 19
C. 26
D. 54

50. A spacecraft at Sun-Earth L1 continuously observes the Sun. Which Earth system benefits most directly from early warning of solar eruptions?

- A. Satellite operations and space-weather services
B. Plate tectonics forecasts
C. Ocean salinity maps only
D. Lunar crater dating

ANSWER KEYS

Mock Paper 1

1. D	2. D	3. D	4. A	5. A	6. D	7. A	8. A	9. D	10. A
11. B	12. C	13. C	14. C	15. B	16. B	17. D	18. B	19. A	20. A
21. B	22. B	23. A	24. A	25. C	26. D	27. A	28. B	29. C	30. C
31. C	32. D	33. B	34. D	35. D	36. B	37. C	38. A	39. C	40. B
41. C	42. C	43. D	44. A	45. D	46. C	47. B	48. B	49. A	50. B

Answer distribution: A=13, B=13, C=12, D=12

Mock Paper 2

1. B	2. A	3. A	4. B	5. B	6. B	7. A	8. C	9. D	10. A
11. A	12. A	13. B	14. B	15. A	16. B	17. B	18. D	19. C	20. C
21. C	22. D	23. D	24. A	25. B	26. D	27. C	28. A	29. D	30. C
31. A	32. D	33. D	34. D	35. D	36. A	37. B	38. A	39. B	40. C
41. D	42. D	43. C	44. C	45. B	46. C	47. A	48. C	49. B	50. C

Answer distribution: A=13, B=13, C=12, D=12

Mock Paper 3

1. A	2. D	3. B	4. A	5. D	6. B	7. C	8. B	9. A	10. C
11. B	12. B	13. B	14. D	15. A	16. B	17. B	18. D	19. D	20. A
21. D	22. A	23. A	24. A	25. D	26. C	27. A	28. B	29. C	30. D
31. D	32. C	33. D	34. A	35. B	36. C	37. A	38. B	39. C	40. A
41. D	42. C	43. D	44. C	45. C	46. B	47. B	48. C	49. A	50. C

Answer distribution: A=13, B=13, C=12, D=12

Mock Paper 4

1. C	2. A	3. B	4. D	5. D	6. A	7. D	8. A	9. B	10. B
11. D	12. A	13. B	14. D	15. A	16. A	17. B	18. A	19. A	20. D
21. D	22. A	23. B	24. C	25. A	26. B	27. C	28. C	29. C	30. A
31. B	32. A	33. B	34. D	35. B	36. C	37. D	38. C	39. D	40. B
41. A	42. B	43. C	44. C	45. B	46. C	47. D	48. D	49. C	50. C

Answer distribution: A=13, B=13, C=12, D=12

Mock Paper 5

1. A	2. A	3. B	4. A	5. C	6. A	7. D	8. D	9. A	10. A
11. A	12. B	13. C	14. D	15. D	16. B	17. D	18. B	19. B	20. D
21. D	22. D	23. D	24. D	25. C	26. C	27. B	28. C	29. C	30. D
31. A	32. C	33. D	34. B	35. B	36. C	37. C	38. B	39. C	40. A
41. B	42. A	43. B	44. C	45. B	46. A	47. C	48. A	49. B	50. A

Answer distribution: A=13, B=13, C=12, D=12

OLYMPIAD CHALLENGE - EXPLANATIONS

Mock Paper 1: Questions 41-50

41. **C** - $d = 1/p = 8.00$ pc. Multiplying by 3.26 gives about 26.1 light-years.
42. **C** - Collecting area is proportional to diameter squared: $(40/20)^2 = 4$.
43. **D** - $T = a^{3/2} = 4^{3/2} = 8$ years.
44. **A** - Circular speed varies as $1/\sqrt{r}$; increasing radius by 4 reduces speed by 2.
45. **D** - At equal temperature, luminosity is proportional to surface area and hence R^2 .
46. **C** - Time = distance/speed = $3.6 \times 10^7 / 3 \times 10^8 = 0.12$ s.
47. **B** - Net force = $75,000 - 50,000 = 25,000$ N; $a = 25,000/5000 = 5$ m/s².
48. **B** - Using $1/f = 1/v + 1/u$ with magnitudes: $1/20 = 1/v + 1/60$, so $1/v = 1/30$.
49. **A** - Mean = $(1.00+1.02+0.99+1.01)/4 = 1.005$ s.
50. **B** - Solar plasma and magnetic fields can disturb the magnetosphere and technological systems.

Mock Paper 2: Questions 41-50

41. **D** - $d = 1/p = 5.00$ pc. Multiplying by 3.26 gives about 16.3 light-years.
42. **D** - Angular resolution is proportional to wavelength/aperture.
43. **C** - $T = a^{3/2} = 9^{3/2} = 27$ years.
44. **C** - Escape speed varies as $\sqrt{M/R}$; $\sqrt{4} = 2$.
45. **B** - A much larger surface area can raise total luminosity despite lower surface temperature.
46. **C** - The field is 5 pixels by 5 pixels, so about 25 pixels.
47. **A** - Weight is 2.0×10^6 N. Net thrust is 1.0×10^6 N, so $a = 5$ m/s².
48. **C** - $\lambda = c/f = 3 \times 10^8 / 1.5 \times 10^8 = 2$ m.
49. **B** - The readings agree closely with one another but are far from the accepted value.
50. **C** - Their scientific targets and observing locations differ.

Mock Paper 3: Questions 41-50

41. **D** - $d = 1/p = 20.00$ pc. Multiplying by 3.26 gives about 65.2 light-years.
42. **C** - Area scales as D^2 , so doubling diameter gives four times the photons.
43. **D** - $T = a^{3/2} = 16^{3/2} = 64$ years.
44. **C** - Escape speed varies as $\sqrt{M/R}$; reducing R to $R/4$ multiplies speed by 2.
45. **C** - Stefan-Boltzmann relation gives L proportional to T^4 for equal radius: $2^4 = 16$.
46. **B** - 24 h = 1440 min; $1440/15 = 96$ min.
47. **B** - Thrust = mass flow rate $\times$ exhaust speed = $20 \times 2500 = 50,000$ N.
48. **C** - Magnetic force magnitude qvB is directly proportional to speed.
49. **A** - A straight line through the origin means direct proportionality between the plotted quantities.
50. **C** - Sample return needs ascent, transfer, re-entry and protected recovery.

Mock Paper 4: Questions 41-50

41. **A** - $d = 1/p = 2.50$ pc. Multiplying by 3.26 gives about 8.1 light-years.
42. **B** - Resolution angle is inversely proportional to aperture: $1.2/3 = 0.4$ arcsecond.
43. **C** - $T = a^{3/2} = 25^{3/2} = 125$ years.
44. **C** - Kepler's third law gives T proportional to $r^{3/2}$; $9^{3/2} = 27$.
45. **B** - $v = c(1/500) = 3 \times 10^5$ km/s $\times$ $0.002 = 600$ km/s.
46. **C** - Distance = $ct = 3.0 \times 10^8 \times 0.070 = 2.1 \times 10^7$ m = $21,000$ km.
47. **D** - Initial mass = 10 t and final mass = 2 t, so ratio = 5 .
48. **D** - Net power gained = absorbed - emitted = 150 W.
49. **C** - Temperature rises from 20 to 45 deg C: $25/4 = 6.25$ deg C per minute.
50. **C** - Sustainable operations require tracking, coordination and disposal to reduce collision and debris risk.

Mock Paper 5: Questions 41-50

41. **B** - $d = 1/p = 12.50$ pc. Multiplying by 3.26 gives about 40.8 light-years.
42. **A** - Transmitted photons = $0.60 \times 5000 = 3000$.
43. **B** - $T = a^{3/2} = 36^{3/2} = 216$ years.
44. **C** - $v_{\text{escape}} = \sqrt{2} v_{\text{circular}}$, so $v_{\text{circular}} = 11.2/\sqrt{2}$ about 7.9 km/s.
45. **B** - A shift to longer wavelength is a redshift, commonly associated with recession.
46. **A** - $(80-20)/(80+20) = 60/100 = 0.60$.

- 47. C** - Using $v^2 = u^2 + 2as$: $0 = 2500 - 2000a$, so $a = 1.25 \text{ m/s}^2$.
- 48. A** - Photon energy is inversely proportional to wavelength.
- 49. B** - After ordering, the middle value is 19; the large outlier does not change the median much.
- 50. A** - Solar monitoring supports forecasts of conditions that can affect satellites, radio and power systems.

SYLLABUS COVERAGE AUDIT

Unit	Topic	Paper 1	Paper 2	Paper 3	Paper 4	Paper 5
1	Fundamentals of Astronomy	6	6	6	6	6
2	The Solar System	5	5	5	5	5
3	Stars, Galaxies & the Universe	5	5	5	5	5
4	Earth & Moon	4	4	4	4	4
5	Satellites & Space Technology	5	5	5	6	5
6	Rockets & Launch Vehicles	6	6	6	5	6
7	ISRO & India's Space Missions	4	5	5	4	5
8	Space Exploration	4	4	4	4	4
9	Basic Physics for Space Science	5	5	5	5	5
10	Scientific Reasoning & Logical Aptitude	3	3	3	3	3
11	General Space Awareness	3	2	2	3	2

Every paper includes questions from all eleven units. The final ten questions in each paper are the designated difficult Olympiad Challenge questions.

Current-Mission Source Notes

- ISRO: Aditya-L1 mission pages and July 2026 observing announcement - operational solar observations from Sun-Earth L1.
- ISRO: XPoSat mission pages and June 2026 observing announcement - X-ray polarimetry and spectroscopy of cosmic sources.
- ISRO: July 2026 Integrated Main Parachute Air Drop Test - Gaganyaan crew-module descent-system qualification.
- ISRO: approved Chandrayaan-4 and Venus Orbiter Mission information - future lunar sample-return and Venus science objectives.
- NASA: Artemis II official 2026 mission reporting - crewed flight around the Moon.
- NASA and international guidance: commercial low-Earth-orbit development and space-sustainability principles.

Information cut-off for time-sensitive questions: 23 July 2026. Future missions are identified as planned, approved or under development rather than presented as completed.

Teacher Review Checklist

- All questions are single-correct MCQs.
- Each paper contains exactly 50 questions.
- Questions 41-50 in every paper are deliberately difficult.
- All eleven syllabus units appear in every paper.
- Answer keys were generated directly from the final option order.
- Numerical challenge answers were recalculated programmatically.